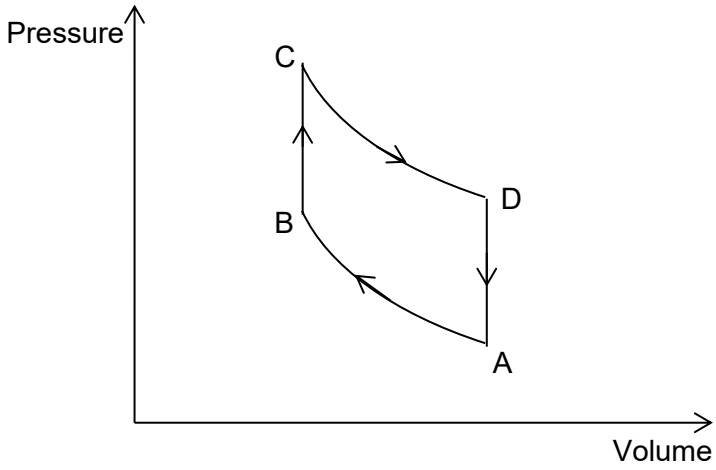
4(a) State the First Law of Thermodynamics in words. [1]

(b) A gas in an engine undergoes a cyclic process ABCD, as shown in the pressure – volume diagram below:



AB and CD are adiabatic processes, while BC and DA are isochoric processes.

(i) For process AB, the work done is 10 J. Calculate the change in internal energy of the gas. State whether it is an increase or decrease. [2]

For process BC, 70 J of heat is supplied to the gas, while for process CD, 25 J of work is done.

(ii) Calculate the heat flow during process DA. State whether it is into or out of the gas. [3]

(iii) Calculate the efficiency of the engine. [2]

(iv) The engine executes 3000 cycles per minute. Calculate its useful power output. [2]